

COMPENG 2DX3
Microprocessor Systems Project
Final Report - ToF Spatial Scanner

Yatukulan Vivekanandan (vivekany)
Student Number: 400566562

Bus Speed Assignment (LSD = 2): 18 MHz
LED Assignment (2nd LSD = 6): PF0 / PF4 / PN1

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As a future member of the engineering profession, the student is responsible for performing the required work in an honest manner, without plagiarism and cheating. Submitting this work with my name and student number is a statement and understanding that this work is my own and adheres to the Academic Integrity Policy of McMaster University and the Code of Conduct of the Professional Engineers of Ontario. Submitted by [Yatukulan Vivekanandan, vivekany, 400566562]

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1.0 Device Overview

1.1 Features

- 360° planar distance scanning using VL53L1X Time-of-Flight (ToF) sensor
- 32 angular measurements per scan, spaced 11.25° apart
- Up to 25 discrete scan positions along the x-axis (10 cm fixed displacement each)
- I²C communication between MSP432E401Y microcontroller and ToF sensor at 100 kbps
- UART serial transmission to PC at 115,200 baud for real-time data streaming
- 28BYJ-48 stepper motor driven via ULN2003 for precise angular positioning
- Individualized bus speed: 18.461 MHz (PSYSDIV = 25)
- Three onboard status LEDs with distinct roles (PF0, PF4, PN1)
- Two push buttons for system start/stop (PJ0) and scan trigger (PJ1)
- CSV-formatted output compatible with MATLAB 3D reconstruction
- Polling-based design with edge detection for reliable button response

1.2 General Description

This spatial mapping system is a custom embedded LIDAR-equivalent platform designed for indoor environment mapping. The system mounts a VL53L1X ToF sensor on a 28BYJ-48 stepper motor, enabling a full 360° sweep in the y-z plane. At each of 32 angular positions (every 11.25°), a distance measurement is taken, and all data is immediately streamed to a connected PC over UART in CSV format.

The user physically repositions the unit along the x-axis between scans (every 10 cm), pressing a dedicated scan button each time. Up to 25 scan planes can be collected, producing a full 3D point cloud of the surveyed space. The data is post-processed in MATLAB to generate a 3D wireframe or point-cloud visualization.

The system runs on the MSP432E401Y microcontroller, configured at an individualized bus speed of 18.461 MHz. Three onboard LEDs provide real-time feedback: one flashes on each ToF measurement, one signals UART transmission blocks, and one indicates stepper motor activity.

1.3 Block Diagram

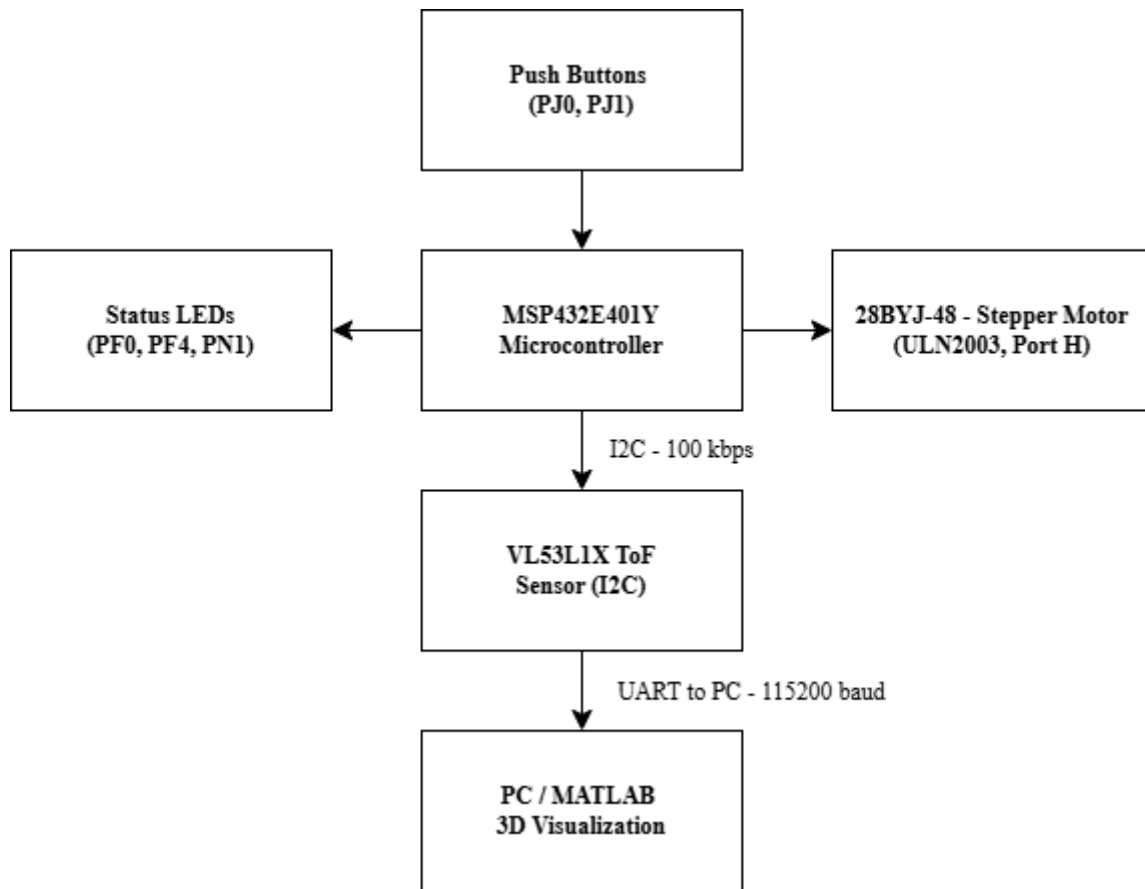


Figure 1: System data flow block diagram. The microcontroller orchestrates all subsystems: reading buttons, driving the stepper motor, polling the TOF sensor, and sending data over UART, and updating the status LEDs.

2.0 Device Characteristics Table

Table 1: Key system characteristics and operating parameters

Parameter	Value	Notes
System Bus Clock	18.461 MHz	PSYSDIV 25 - 480MHz/(25+1)
I ² C Clock	100 kbps	Standard Mode
UART Baud Rate	115200 bps	Communication to PC
Stepper Steps	2048 steps	28BYJ-48 full-step drive
Angular Resolution	11.25°	2048/32 = 64 steps/angle
Number of Angles/Scans	32	Full 360° per plane
Max Scan Planes (x-axis)	25	Manual 10 cm displacement

Displacement per Step	10 cm	User-repositioned
ToF Measurement Range	0-4000mm	Maximum Range of ToF sensor
Step Delay (min)	10 ms/step	Prevents motor skipping
Distance LED (Measurement)	PF0 (D4)	Blinks when a distance is captured
UART Tx LED	PF4 (D3)	Blinks when sending data to PC
Motor Active LED	PN1 (D1)	ON while stepper is running
Start/Stop Button	PJ0 (SW1)	Toggles system state
Scan Trigger Button	PJ1 (SW2)	Initiates one 360° scan

3.0 Detailed Description

3.1 Distance Measurement

3.1.1 ToF Sensor Integration

The VL53L1X sensor communicates with the microcontroller over I²C (PB2 = SCL, PB3 = SDA) at 100 kbps. The sensor's default I2C address is 0x29. On power-up, the firmware polls the sensor's boot-complete flag (VL53L1X BootState) before calling VL53L1X SensorInit and VL53L1X StartRanging to enter continuous measurement mode.

At each angular position, ToF GetDistance() polls the sensor until dataReady = 1, then reads range status and distance. Readings are validated: rangeStatus == 0 and distance within 0–4000 mm. Invalid readings fall back to the last known good distance (lastValidDist, initialized to 500 mm).

3.1.2 Stepper Motor and Angular Positioning

The 28BYJ-48 stepper motor is driven in full-step mode via a ULN2003 driver board connected to PH3–PH0. One full revolution requires 2048 steps. Each of the 32 scan angles is 11.25° apart, requiring $2048/32 = 64$ steps per advance. The four-step drive sequence energizes two coils simultaneously for maximum torque:

$$fullStepSeq = \{0x03, 0x06, 0x0C, 0x09\}$$

After all 32 measurements, the motor returns home by rotating $32 \times 64 = 2048$ steps counter-clockwise (exactly one full revolution). Coils are de-energized after each rotation to prevent heat buildup.

3.1.3 Data Acquisition Sequence

Each call to PerformScan(scanIndex) executes the following sequence:

1. Turn on motor-active LED (PN1).
2. Flash UART Tx LED (PF4) to signal start of data block.
3. Clear any stale ToF interrupt.
4. For each of 32 angles:
 - a. Read ToF distance.
 - b. Flash measurement LED (PN0).

- c. Transmit CSV record over UART: scanIndex, angle (deg), displacement (mm), distance (mm).
- d. Rotate stepper 64 steps clockwise.
5. Return stepper to home (2048 steps CCW).
6. Turn off motor-active LED.
7. Flash UART Tx LED to signal end of data block.

3.1.4 UART Data Format

Each measurement is transmitted as a comma-separated line:

`<scanIndex>,<angle>,<displacement mm>,<distance mm>\r\n`

Example: 0, 11.25, 0, 1450\r\n - scan 0, angle 11.25°, 0 mm displacement, 1450 mm distance.

3.2 Visualization

3.2.1 MATLAB Data Parsing and Coordinate Transformation

The UART output from the microcontroller is read by a MATLAB script on the PC. Each CSV record is parsed to extract the polar angle θ , the axial displacement x , and the radial distance r . Cartesian coordinates are computed as: $y = r \cos(\theta)$, $z = r \sin(\theta)$. The coordinate x is determined from the scan index and fixed displacement between scans, allowing each full rotation to be placed at a different position along the x -axis.

3.2.2 Embedded Data Transmission

On the embedded side, the firmware generates this data stream in real time. During each call to PerformScan(), the system measures distance at 32 angular positions and immediately transmits each result over UART in CSV format, ensuring a continuous and ordered flow of data to MATLAB.

In MATLAB, the converted (x, y, z) coordinates are stored and visualized dynamically. The *scatter3* function is used to plot all measurement points as a 3D point cloud, providing a direct representation of the scanned environment. To enhance spatial structure, the script also uses *plot3* to connect adjacent points within each scan, forming circular rings that represent a full 360° sweep. Furthermore, corresponding points between consecutive scans are connected with dashed lines, creating vertical links between rings.

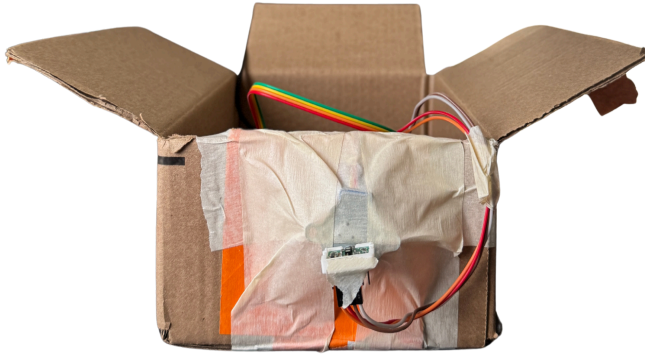
This produces a 3D wireframe model that evolves in real time as new scans are completed, ultimately reconstructing the geometry of the surrounding environment.

4.0 Application Note, Instructions, and Expected Output

4.1 Hardware Setup

1. Connect the 28BYJ-48 stepper motor to the ULN2003 driver board; connect IN1–IN4 to PH0–PH3 on the LaunchPad.
2. Connect VL53L1X breakout board:
 - a. SDA → PB3, SCL → PB2, VCC → 3.3V, GND → GND.

3. Mount the ToF sensor vertically on the stepper motor shaft so the sensing direction lies in the y-z plane.
4. Ensure the USB UART cable is connected (PA0/PA1 or onboard USB).
5. No external buttons or LEDs are required - onboard SW1 (PJ0), SW2 (PJ1), D1 (PN1), D3 (PF4), and D4 (PF0) are used.



(a) Front View



(b) Top Down View

Figure 2: Front and Top Down View of the Spatial Scanner

4.2 Software Setup and Flashing

1. Open the project in Keil μ Vision with MSP432E401Y.
2. Ensure all driver files are present: PLL.c, SysTick.c, uart.c, onboardLEDs.c, VL53L1X api.c.
3. Build and flash to the MSP432E401Y LaunchPad.
4. Open a serial terminal at 115,200 baud, 8N1.

4.3 Operating Procedure

1. Power on the LaunchPad. The system will print initialization messages and wait for the ToF sensor to boot.
2. Position the scanner at the first measurement location ($x = 0$ cm).
3. Press PJ0 (SW1) to start the system. The terminal will confirm: System STARTED.
4. Press PJ1 (SW2) to trigger Scan 1. The system will:
 - a. Turn on PN1 (D1) - motor is active.
 - b. Flash PF4 (D3) - UART block beginning.
 - c. Rotate 360° in 32 steps, transmitting one CSV line per angle, flashing PN0 (D2) at each.
 - d. Return to home, turn off PN1, flash PF4 again - block complete.
5. Move the scanner 10 cm along the x-axis. Press PJ1 again for Scan 2. Repeat up to 25 times.

6. After all desired scans, press PJ0 to stop the system, or wait for the 25-scan quota to complete automatically.
7. Save the terminal log as scan data.csv and run the MATLAB visualization script.

4.4 Expected Output

```

=== 2DX3 Project Deliverable 2 - System Ready ===
Student: 400566562 | Bus: 18.461 MHz (PSYSDIV=25)
LEDs: PF0=Measurement | PF4=UART_Tx | PNI=MotorActive
PJ0=Start/Stop | PJ1=Trigger Scan
ToF Sensor ready.
System STARTED. Press PJ1 to begin scan 1.
Scan 1/25 starting (disp=0 mm)...
0,0.00,0,1450
0,11.25,0,1432
.....
0,22.50,0,1405
Move 10 cm and press PJ1 for scan 2.

```

Figure 3: Realterm Output

```

=== 2DX3 Deliverable 2 | Yathu Vivekanandan ===
25 scans total, 32 angles each (11.25 deg apart)
Press PJ0 to START, then PJ1 for each scan.
Move board 10 cm between each PJ1 press.

Press Enter in MATLAB when board is ready...

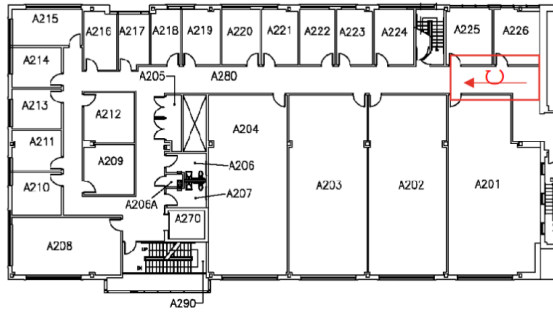
Listening for scan data...

--- Waiting for scan 1/25 ---
Angle 1/32: 0.00 deg -> 2361 mm
Angle 2/32: 11.25 deg -> 2534 mm
Angle 3/32: 22.50 deg -> 2684 mm
Angle 4/32: 33.75 deg -> 256 mm
Angle 5/32: 45.00 deg -> 201 mm

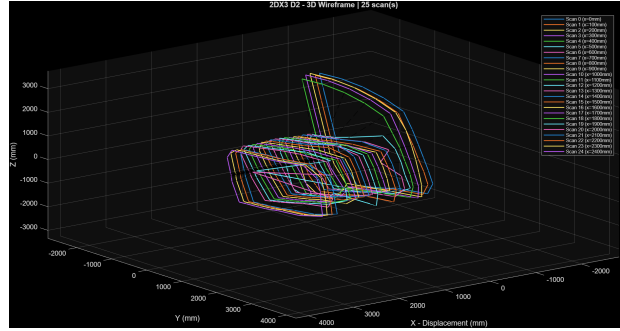
```

Figure 4: MATLAB Expected Output in Command Window

4.4 Campus Location Scan



(a) Assigned Campus Location
(ITB 2nd Floor)



(b) 3D MATLAB model of
ITB 2nd Floor

Figure 5: Side by side comparison of ITB 2nd Floor scanning location and the corresponding 3D reconstruction using the Spatial Scanner.

5.0 Limitations

5.1 Floating-Point and Trigonometric Capability of the MSP432E401Y

The MSP432E401Y uses a single-precision (32-bit) floating-point unit, which is sufficient for basic arithmetic operations such as angle calculations in firmware. However, more complex trigonometric functions introduce additional computational overhead. Due to these constraints and the lack of double-precision support, performing trigonometric calculations on the microcontroller can reduce efficiency and accuracy. As a result, all trigonometric and coordinate transformation operations are offloaded to MATLAB on the PC, avoiding real-time processing limitations on the embedded system.

5.2 Maximum Quantization Error for the ToF Module

The VL53L1X distance measurement is represented with 1 mm resolution, so the maximum quantization error is ± 0.5 mm. This is the error caused by converting a continuous physical distance into discrete 1 mm output steps.

$$\text{Maximum quantization error} = \pm \text{resolution} / 2$$

$$\begin{aligned} & \pm 1 \text{ mm} / 2 \\ & = \pm 0.5 \text{ mm} \end{aligned}$$

5.3 Maximum UART Rate and Implementation

The maximum standard serial communication rate used for communication with the PC was 115200 baud, and this is the rate implemented in the project. Verification was performed by successfully receiving correctly formatted CSV data records in MATLAB in

real time during scanning, which confirmed reliable UART communication between the microcontroller and PC.

5.4 ToF Microcontroller Communication

The VL53L1X ToF sensor communicates with the microcontroller using the I2C protocol (I2C0) over pins PB2 (SCL) and PB3 (SDA). The bus was configured for a communication speed of 100 kbps (standard mode). This speed was set using the I2C timer prescaler register (I2C0_MTPR = 0x08), based on the system clock of 18.461 MHz. This configuration provides reliable communication while meeting the timing requirements of the sensor.

5.5 Primary Speed Bottleneck

The primary limitation on overall system speed is the stepper motor delay. In the firmware, each motor step includes a programmed delay of 10 ms, and one scan uses 32 angular positions with 64 steps per position, giving 2048 clockwise steps during data acquisition. After the scan is complete, the motor returns to its home position using another 2048 counterclockwise steps. Therefore, one full scan cycle requires 4096 total motor steps, corresponding to a minimum motor time of $4096 \times 10 \text{ ms} = 40.96 \text{ s}$. By comparison, the ToF measurement delay is much smaller, since the sensor is polled in 5 ms intervals and only 32 measurements are taken per scan, while UART transmission at 115200 baud adds only a small additional delay. This shows that the stepper motor timing, rather than sensing or communication, is the dominant limitation on system speed.

5.6 Bus Speed Configuration

The system clock is configured using the PLL with the formula:

$$\begin{aligned} \text{System Clock} &= 480 \text{ MHz} / \text{PSYSDIV} + 1 \\ \text{PSYSDIV} &= 25 \end{aligned}$$

$$\begin{aligned} \text{System Clock} &= 480 / 26 \\ &= 18.461 \text{ MHz} \end{aligned}$$

This matches the assigned bus speed from Table 1 (LSD = 2 $\rightarrow$ 18 MHz), with the exact value determined by the PLL configuration.

5.7 Status LED Meanings

The second least significant digit of student number 400566562 is 6. From Table 2 of the specification, the LED assignments are:

Table 2: Status LED assignments (2nd LSD = 6)

LED	GPIO	Meaning
Measurement Status	PF0 (D4)	Flashes on each ToF Distance Measurement
UART Tx Status	PF4 (D3)	Flashes at the start and end of each UART CSV data block
Additional Status	PN1 (D1)	ON while stepper motor is spinning; OFF otherwise

6.0 Circuit Schematic

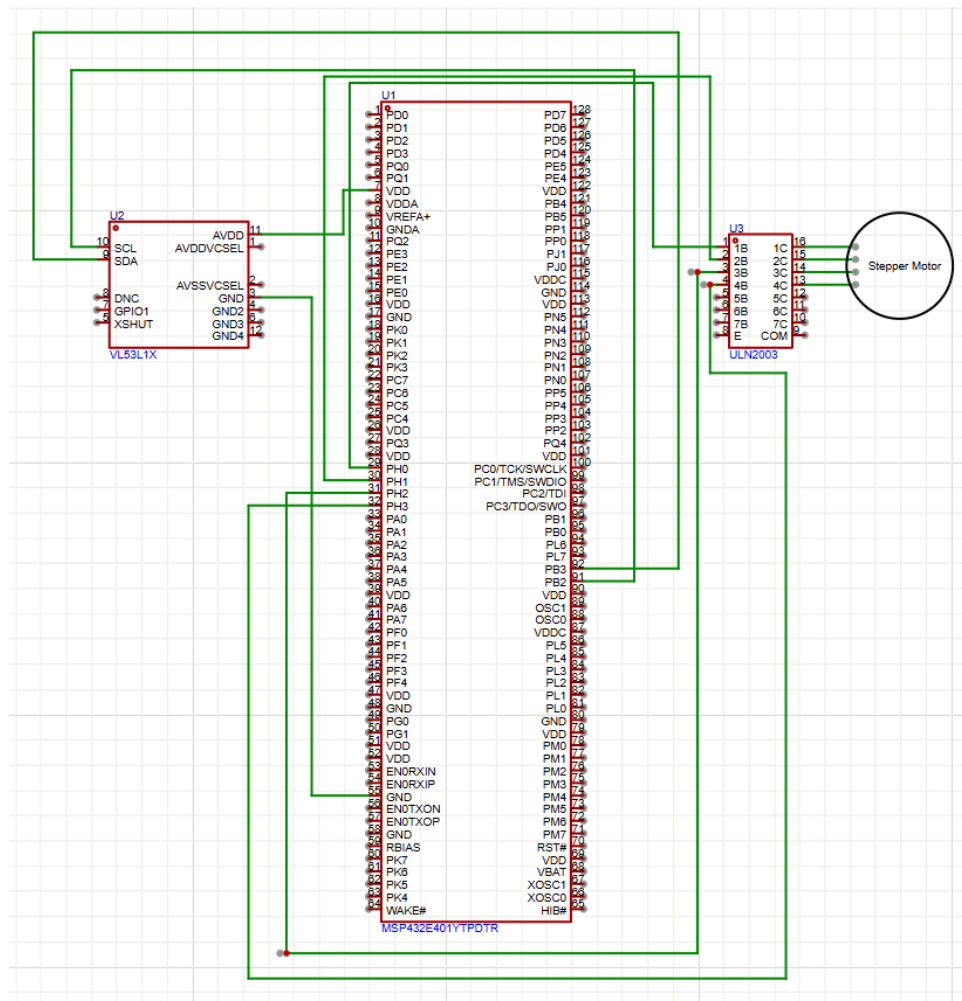


Figure 6: Circuit Schematic using EasyEDA

7.0 Programming Logic Flowcharts

7.1 Keil Code Flowchart

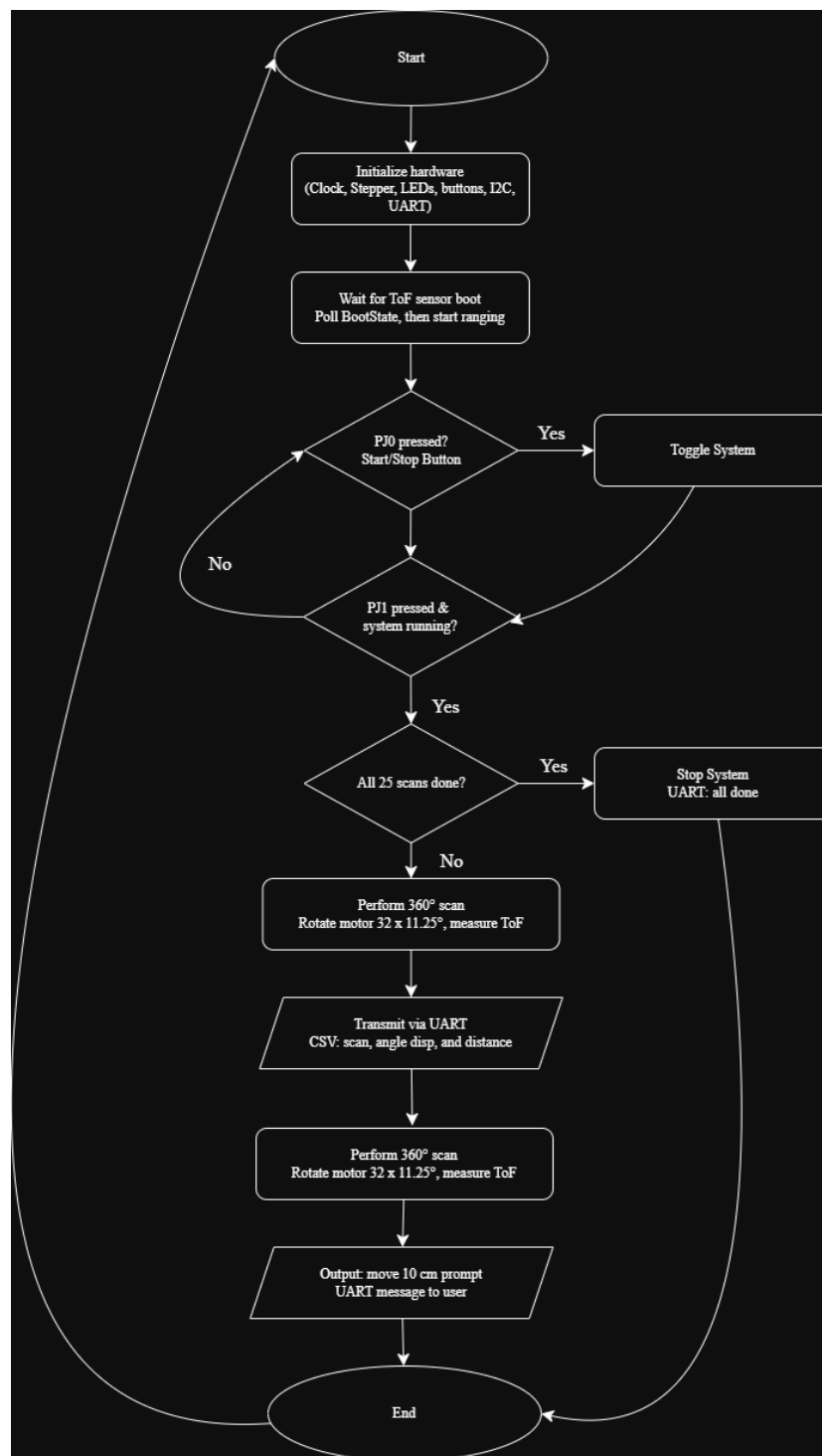


Figure 7: Keil Code Flowchart

7.2 MATLAB Code Flowchart

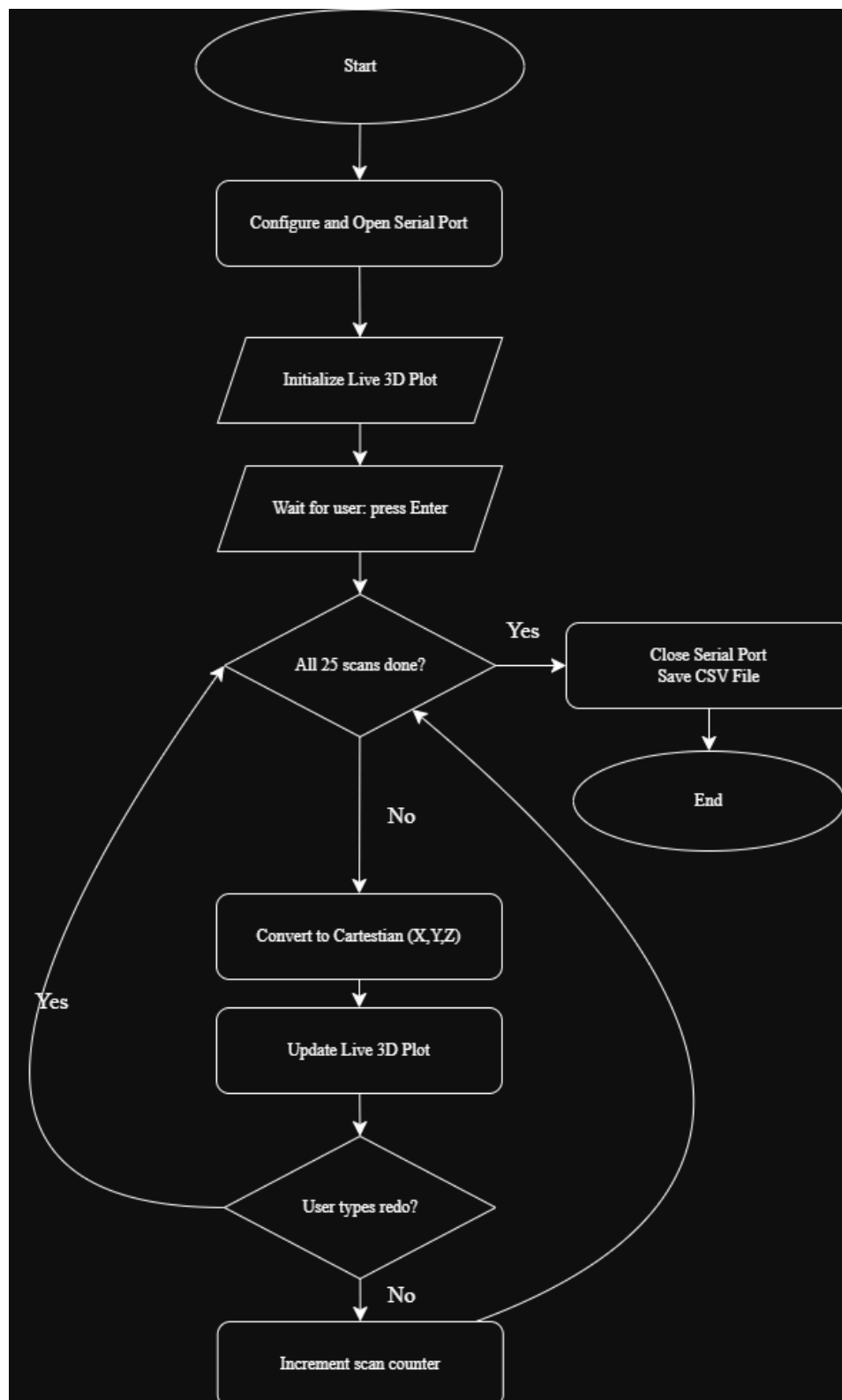


Figure 8: MATLAB Code Flowchart

References

- [1] "2025_2026_2DX3_Laboratory_Manual," class notes for 2DX3, Department of Electrical and Computer Engineering, McMaster University, 2026.